12.163

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Question

The geometric transformation specified by

$$\begin{pmatrix} X' & Y' & 1 \end{pmatrix} = \begin{pmatrix} X & Y & 1 \end{pmatrix} \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

in a 2D CAD system represents

a) Scaling and Translation
b) Scaling and Rotation
c) Rotation and Translation
d) Rotation

Theoretical Solution

A 2D affine transformation is of the form

$$\begin{pmatrix} \mathbf{x}'^{\top} & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{x}^{\top} & 1 \end{pmatrix} \mathbf{T} \tag{1}$$

where the transformation matrix is

$$\mathbf{T} = \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{t}^{\top} & 1 \end{pmatrix}$$

(1) simplifies to

$$\begin{pmatrix} \mathbf{x}'^{\top} & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{x}^{\top} & 1 \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{t}^{\top} & 1 \end{pmatrix} \tag{2}$$

$$= \begin{pmatrix} \mathbf{x}^{\top}\mathbf{A} + \mathbf{t}^{\top} & 1 \end{pmatrix} \tag{3}$$

$$\mathbf{x}'^{\top} = \mathbf{x}^{\top}\mathbf{A} + \mathbf{t}^{\top} \tag{4}$$

$$\mathbf{x}' = \mathbf{A}^{\top}\mathbf{x} + \mathbf{t} \tag{5}$$

Theoretical Solution

The given transformation matrix is:

$$\mathbf{T} = \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \\ 1 & 2 & 1 \end{pmatrix} \tag{6}$$

From this, we identify the linear transformation matrix $\mathbf{A}$ and the translation vector $\mathbf{t}$.

$$\mathbf{A} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix}, \ \mathbf{t} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \tag{7}$$

Since $\mathbf{A}$ is a diagonal matrix and not a multiple of the identity, it represents a non-uniform scaling. Since $\mathbf{t} \neq \mathbf{0}$, there is a translation.

Theoretical Solution

A pure rotation requires the linear part $\mathbf{A}$ to be an orthogonal matrix, where $\mathbf{A}\mathbf{A}^\top = \mathbf{I}$. We check this condition:

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} = \begin{pmatrix} 0.25 & 0 \\ 0 & 0.0625 \end{pmatrix} \neq \mathbf{I} \tag{8}$$

Since $\mathbf{A}$ is not orthogonal, the transformation is not a rotation.

Example

Applying the transformation to the point $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. In homogeneous coordinates, using (5)

$$\mathbf{x}' = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \tag{9}$$

$$= \begin{pmatrix} 0.5 \\ 0.25 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \tag{10}$$

$$= \begin{pmatrix} 1.5 \\ 2.25 \end{pmatrix} \tag{11}$$

The point $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is transformed to $\begin{pmatrix} 1.5 \\ 2.25 \end{pmatrix}$, demonstrating both scaling and translation.

The correct option is **a) Scaling and Translation**.

Plot

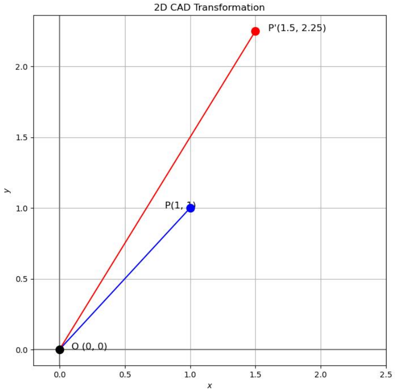

Figure: Plot